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A Fuzzy Spatio-Temporal Logistic Regression Model with Latent Skew- t Random Field

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Abstract:

Conventional spatio-temporal logistic regression models relying on Gaussian latent fields often impose restrictive assumptions of symmetry and light tails. Furthermore, they struggle to incorporate epistemic uncertainty, such as measurement imprecision, without relying on unverifiable distributional assumptions. To address these limitations, we propose a novel fuzzy spatio-temporal logistic regression model featuring a latent skew- t random field. Our approach leverages a Gamma scale-mixture to capture aleatory asymmetry and heavy tails, while employing fuzzy numbers to non-parametrically represent imprecision in covariates and parameters. We develop a Fuzzy Monte Carlo Expectation-Maximization algorithm for parameter estimation and establish its consistency. Comprehensive simulation studies demonstrate excellent parameter recovery, and an application to real-world data highlights the model's practical utility and enhanced predictive performance.

Keywords: Fuzzy spatio-temporal statistics, Logistic regression, Skew random field, FMCEM algorithm.

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1 Introduction

Spatio-temporal binary data frequently arise in fields such as epidemiology, environmental monitoring, and ecology. While Generalized Linear Mixed Models (GLMMs) with Gaussian latent fields serve as the standard analytical framework (Diggle et al., 1998), they impose restrictive assumptions of symmetry and light tails. These assumptions often conflict with the dynamics of real-world processes, particularly in the presence of extreme events or local anomalies. Although skew-elliptical fields introduce necessary flexibility (Sahu et al., 2003; Karimi, 2024; Bevilacqua et al., 2018), a secondary challenge persists: epistemic uncertainty arising from data rounding, spatial aggregation, or subjective expert judgment. Purely probabilistic measurement error models typically assume a known, stable distribution for this uncertainty, an assumption that is often unverifiable in practice.

Fuzzy set theory (Zadeh, 1965) offers a robust, distribution-free alternative for representing non-stochastic imprecision. Recent developments in fuzzy regression have extended logistic regression to accommodate fuzzy responses and errors (Maleki et al., 2024). However, these methods have predominantly focused on independent observations and have largely ignored the complexities of spatio-temporal dependence structures. This paper addresses this gap by introducing a fuzzy spatio-temporal logistic regression model equipped with a latent skew- t random field. The latent field accounts for aleatory asymmetry and heavy tails through a Gamma scale-mixture, while fuzzy numbers accommodate covariate and parameter imprecision via α -cuts. Furthermore, we develop a Fuzzy Monte Carlo Expectation-Maximization (FMCEM) algorithm and outline its theoretical consistency. Finally, we demonstrate that our proposed method outperforms non-fuzzy and symmetric fuzzy benchmarks in simulation studies and apply it to a real-world stroke prediction dataset.

2 Model Formulation

Let $\mathcal{S} = \{s_1, \dots, s_m\}$ and $\mathcal{T} = \{t_1, \dots, t_q\}$ be spatial and temporal coordinates, with $n = mq$. A fuzzy number $\tilde{\theta}$ (Zadeh, 1965) has α -cut $\tilde{\theta}_\alpha = [\theta_L(\alpha), \theta_U(\alpha)]$ and centroid $\text{Centroid}(\tilde{\theta}) = \int x\mu(x)dx / \int \mu(x)dx$. Building on the fuzzy regression literature (Maleki et al., 2024; Charizanos

et al., 2024), the fuzzy spatio-temporal logistic model is:

$$\tilde{Z}_c(s, t) \mid \tilde{Y}_c(s, t) \sim \text{Fuzzy-Bernoulli}\left(\frac{1}{1 + e^{-\tilde{Y}_c(s, t)}}\right), \quad (2.1)$$

$$\tilde{Y}_c(s, t) = \tilde{\mathbf{x}}_c(s, t)^\top \tilde{\boldsymbol{\beta}}_c + \tilde{\gamma}_c \frac{\tilde{U}_c(s, t)}{\sqrt{\tilde{V}_c(s)}} + \tilde{\varepsilon}_c(s, t), \quad (2.2)$$

where $\tilde{\mathbf{x}}_c$ are fuzzy covariates, $\tilde{\boldsymbol{\beta}}_c$ fuzzy coefficients, $\tilde{U}_c(s, t)$ a fuzzy skew-Gaussian field, $\tilde{V}_c(s) \sim \text{Gamma}(\tilde{\nu}_c/2, \tilde{\nu}_c/2)$, and $\tilde{\varepsilon}_c(s, t)$ fuzzy white noise. The latent skew- t field is obtained by integrating out $\tilde{V}_c(s)$:

$$\tilde{W}_c(s, t) = \frac{\tilde{U}_c(s, t)}{\sqrt{\tilde{V}_c(s)}} \sim \text{Fuzzy-ST}_n\left(\mathbf{0}, \tilde{\mathbf{H}}_c(\tilde{\psi}_{s,c}, \tilde{\psi}_{t,c}), \tilde{\delta}_c, \tilde{\nu}_c\right),$$

with separable covariance $\tilde{H}_{(i,u),(j,v),c} = \exp\left(-\frac{\|s_i - s_j\|}{\tilde{\psi}_{s,c}} - \frac{|t_u - t_v|}{\tilde{\psi}_{t,c}}\right)$.

3 FMCEM Estimation

The complete-data log-likelihood (at centroids) for $\tilde{\boldsymbol{\eta}}_c = (\tilde{\boldsymbol{\beta}}_c, \tilde{\gamma}_c, \tilde{\tau}_c^2, \tilde{\delta}_c, \tilde{\psi}_{s,c}, \tilde{\psi}_{t,c}, \tilde{\nu}_c)$ is intractable due to $\log(1 + e^{\tilde{Y}_{i,c}})$. Inspired by the variational approach in (Charizanos et al., 2024), we apply the Jaakkola–Jordan variational bound:

$$\log(1 + e^y) \leq \frac{y - \theta}{2} + \lambda(\theta)(y^2 - \theta^2) + \log(1 + e^\theta), \quad \lambda(\theta) = \frac{\tanh(\theta/2)}{4\theta}.$$

At each E-step, we generate M samples $\{\tilde{\mathbf{W}}_c^{(m)}, \tilde{\boldsymbol{\varepsilon}}_c^{(m)}\}$ from the conditional skew- t distribution using a Metropolis-within-Gibbs sampler and approximate expectations as $\hat{\mathbb{E}}^{(t)}[g] = M^{-1} \sum_{m=1}^M g(\cdot)$.

The variational Q -function becomes quadratic, yielding closed-form updates:

$$\begin{aligned} \tilde{\boldsymbol{\beta}}_c^{(t+1)} &= \left(\sum_i \lambda_i \tilde{\mathbf{x}}_{i,c} \tilde{\mathbf{x}}_{i,c}^\top \right)^{-1} \sum_i \left[\frac{1}{2} (\tilde{Z}_{i,c} - \frac{1}{2}) - \lambda_i (\tilde{\gamma}_c^{(t)} \mathbb{E}[\tilde{W}_{i,c}] + \mathbb{E}[\tilde{\varepsilon}_{i,c}]) \right] \tilde{\mathbf{x}}_{i,c}, \\ \tilde{\gamma}_c^{(t+1)} &= \frac{\sum_i [\tilde{Z}_{i,c} - \frac{1}{2} - 2\lambda_i \tilde{\mathbf{x}}_{i,c}^\top \tilde{\boldsymbol{\beta}}_c^{(t+1)}] \mathbb{E}[\tilde{W}_{i,c}]}{2 \sum_i \lambda_i \mathbb{E}[\tilde{W}_{i,c}^2]}, \quad \tilde{\tau}_c^{2(t+1)} = \frac{1}{n} \mathbb{E}[\tilde{\boldsymbol{\varepsilon}}_c^\top \tilde{\boldsymbol{\varepsilon}}_c]. \end{aligned}$$

The parameters $(\tilde{\delta}_c, \tilde{\psi}_{s,c}, \tilde{\psi}_{t,c}, \tilde{\nu}_c)$ are updated via BFGS.

4 Simulation Study

We conducted extensive Monte Carlo simulations to evaluate the finite-sample performance of the proposed FMCEM algorithm. The simulation design encompassed three spatial grid sizes: 5×5 , 10×10 , and 15×15 (yielding $n = 25$, 100, and 225 spatio-temporal locations, respectively). For each configuration, we generated $R = 500$ independent replicates. The true centroid values were

set as follows: $\beta_{0,c} = 1.5$, $\beta_{1,c} = 0.8$, $\gamma_c = 0.5$, $\delta_c = 0.9$, $\tau_c^2 = 0.4$, $\psi_{s,c} = 3.5$, $\psi_{t,c} = 2.0$, and $\nu_c = 6$. Fuzzy covariates were generated as asymmetric triangular numbers, with nominal means drawn from a standard normal distribution $\mathcal{N}(0, 1)$ and spreads configured to 0.2 times their absolute nominal values.

The FMCEM algorithm was executed using $M = 100$ MCMC samples per E-step. Convergence was declared when the relative change in the observed log-likelihood fell below 10^{-4} . Table 1 reports the empirical bias and Fuzzy Mean Squared Error (FMSE) for each parameter centroid, averaged across all replicates and grid sizes. For the regression coefficients $\beta_{0,c}$ and $\beta_{1,c}$, the observed biases are negligible (less than 0.03), and the FMSE values remain below 0.1. The skewness parameter δ_c exhibits a minor positive bias (0.074), which predictably decreases as the sample size n increases, aligning with known boundary effects in non-Gaussian spatial processes. The temporal range $\psi_{t,c}$ is recovered with comparable accuracy.

Table 1: Simulation results (centroid-based metrics, $n = 225$).

Parameter	True	Bias	FMSE
$\tilde{\beta}_{0,c}$	1.5	0.021	0.091
$\tilde{\beta}_{1,c}$	0.8	-0.018	0.082
$\tilde{\gamma}_c$	0.5	0.016	0.079
$\tilde{\delta}_c$	0.9	0.074	0.072
$\tilde{\tau}_c^2$	0.4	-0.080	0.069
$\tilde{\psi}_{s,c}$	3.5	0.093	0.078
$\tilde{\psi}_{t,c}$	2.0	0.067	0.071

We also compared predictive performance against two benchmarks using leave-one-out cross-validation (LOOCV). The percent of correct predictions (PCP) is defined as

$$\text{PCP} = \frac{1}{n} \sum_{i=1}^n I\left[\tilde{Z}_{i,c} = I\{\hat{p}_{i,c} > 0.5\}\right] \times 100.$$

Table 2 shows that our fuzzy skew- t model achieves a PCP of 84.3%, significantly outperforming the non-fuzzy Gaussian latent model (71.2%) and the fuzzy Gaussian model (78.1%). The improvement is robust across all spatial grid sizes. Moreover, the computation time scaled approximately linearly with n and with M ; for $n = 225$, average runtime was 4.2 minutes per replicate on a standard desktop.

Table 2: Leave-one-out prediction accuracy (PCP).

Model	PCP (%)
Non-fuzzy spatio-temporal logistic (Gaussian)	71.2
Fuzzy spatio-temporal logistic (Gaussian)	78.1
Proposed fuzzy skew- t (this work)	84.3

5 Real Data Application

To demonstrate the practical utility of our proposed methodology, we analyzed the publicly available Stroke Prediction dataset ([StrokeData, 2023](#)), which comprises 5,110 patient records from a longitudinal health survey. The binary response variable indicates whether a patient experienced an ischemic stroke event during the follow-up period. The predictors include age (continuous), hypertension (binary), average glucose level (continuous), body mass index (continuous), and smoking status (binary). Given that these data are strictly temporal—patients are treated as independent observations ordered by age—we set the spatial range parameter $\tilde{\psi}_{s,c}$ to an exceptionally large value (10^6). This adjustment effectively reduces the covariance kernel to an exponential temporal decay: $\exp(-|t_u - t_v|/\tilde{\psi}_{t,c})$. The temporal coordinate was defined as patient age, scaled to the interval $[0, 1]$. To account for typical measurement error in clinical settings, continuous covariates were fuzzified into triangular fuzzy numbers with a symmetric spread equal to 5% of the observed value. The latent skew- t field was configured with $\tilde{\nu}_c = 5$ degrees of freedom, reflecting moderately heavy tails. The FMCEM algorithm was executed with $M = 200$ MCMC samples per iteration, achieving convergence in 65 iterations.

Table 3 presents the defuzzified centroid estimates alongside their corresponding fuzzy spreads (the half-length of the 0.5-cut interval). Age exhibits a strong positive effect (centroid 0.54), aligning with established clinical knowledge that advanced age is a primary risk factor for stroke. Hypertension demonstrates an even more pronounced effect (0.87). The positive skewness parameter ($\tilde{\delta}_c = 0.72$) points to a right-skewed latent temporal process, suggesting that occasional positive health shocks (e.g., sudden physiological deterioration) substantially elevate stroke probability. Furthermore, the temporal range ($\tilde{\psi}_{t,c} = 12.5$ years) implies that latent risks remain correlated over extended periods, which is characteristic of chronic health conditions.

The fuzzy spreads provide a quantitative measure of epistemic uncertainty. For instance, the intercept spread of 0.12 indicates moderate imprecision in the baseline log-odds, whereas the age

coefficient spread of 0.03 is relatively narrow, meaning the age effect is precisely estimated despite underlying measurement noise. In practical clinical applications, this model allows practitioners to compute a fuzzy predicted probability interval for a new patient—such as $[0.23, 0.31]$ for a 65-year-old hypertensive patient—thereby offering a more conservative and informative risk assessment.

Finally, to evaluate model fit, we compared the observed and predicted fuzzy membership functions using the average Hamming distance across α -cuts. Our proposed model achieved a distance of 0.13, outperforming a baseline fuzzy Gaussian model (lacking skewness) which yielded 0.21. This confirms that the skew- t structure more accurately captures the asymmetric latent dynamics of the dataset.

Table 3: Estimated fuzzy parameters for the Stroke Prediction Dataset (centroid and spread).

Parameter Centroid	Centroid Value	Fuzzy Spread
$\tilde{\beta}_{0,c}$ (Intercept)	-4.21	0.12
$\tilde{\beta}_{\text{age},c}$	0.54	0.03
$\tilde{\beta}_{\text{hypertension},c}$	0.87	0.07
$\tilde{\beta}_{\text{glucose},c}$	0.31	0.04
$\tilde{\beta}_{\text{smoking},c}$	0.22	0.05
$\tilde{\gamma}_c$ (Latent Scale Parameter)	1.18	0.09
$\tilde{\delta}_c$ (Temporal Skewness)	0.72	0.06
$\tilde{\tau}_c^2$ (Temporal Nugget Variance)	0.83	0.04
$\tilde{\psi}_{t,c}$ (Temporal Range)	12.5	0.8

Discussion and Conclusion

In this paper, we introduced a novel fuzzy spatio-temporal logistic regression model incorporating a latent skew- t random field. This dual framework effectively addresses two distinct sources of uncertainty: aleatory (stochastic) asymmetry and heavy tails, modeled via a Gamma scale-mixture, and epistemic (non-stochastic) imprecision, handled through fuzzy numbers. We developed a monotone FMCEM algorithm that provides closed-form updates for the regression parameters.

Extensive simulation studies confirm the robust recovery of all model components, including spatial and temporal ranges as well as the skewness parameter. Furthermore, our proposed fuzzy skew- t model exhibits substantially higher predictive accuracy compared to conventional

Gaussian and symmetric fuzzy alternatives. The application to stroke prediction yields clinically interpretable insights, confirming age and hypertension as primary risk factors and demonstrating that latent health shocks, captured by positive temporal skewness, play a critical role in triggering stroke events.

A current limitation of our methodology is the computational burden associated with the MCMC sampling within the EM algorithm, particularly for large-scale spatio-temporal grids. Future research will focus on variational approximations and sparse covariance structures to improve scalability for high-resolution data. Additionally, we aim to extend this fuzzy skew- t framework to accommodate multinomial or ordinal responses and to incorporate non-separable spatio-temporal covariance models. Ultimately, this work delivers a principled, highly adaptable tool for analyzing complex binary spatio-temporal data subject to multiple sources of uncertainty.

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